

# HE-DTimeXer: A short-term electricity price prediction model with exogenous variables

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## ABSTRACT

Existing electricity price forecasting methods suffer from several limitations. These include inadequate modeling of interactive effects between prices and exogenous variables—such as energy, load, and climate—and challenges related to noise and abnormal fluctuations. To overcome these issues, we propose the HE-DTimeXer model. This model integrates the Pearson Correlation Coefficient (PCC) and MIC-TSA for feature selection. It further introduces a Holiday Encoding mechanism to improve prediction accuracy during special periods. Additionally, the model employs a TimeXer architecture with differential attention to enhance robustness against noise. Experiments on multiple real-world datasets show that HE-DTimeXer achieves superior performance, reducing RMSE by 22.81 %–71.74 %, MAE by 17.62 %–78.71 %, and MAPE by 22.01 %–77.09 % compared to deep learning baseline models. The model demonstrates enhanced accuracy in capturing short-term dynamics and holiday-specific patterns, offering robust support for electricity market operations.

## 1. Introduction

With the deepening of China's electricity market reform, power transactions are gradually transitioning toward market-based mechanisms. Regional power grid companies play a key role in market operations. They are responsible for electricity procurement and sales, market settlement, power dispatching, and ensuring the safe and stable operation of the power system. Electricity price serves as a critical reference for these companies in formulating procurement plans and sales strategies. The accuracy of price prediction directly affects their trading decisions. It also influences their market competitiveness and long-term sustainability. However, electricity prices is a direct reflection of supply-demand dynamics [1]. Electricity prices exhibit significant volatility, influenced by both physical factors—such as the materials [2] and equipment used in generation and transmission [3]—and human factors, including carbon pricing, energy generation, and load demand [4]. Therefore, in such a complex and rapidly evolving environment, it has become essential to develop high-accuracy short-term electricity price prediction models. These models must integrate factors such as renewable energy generation, load characteristics, and meteorological conditions. Their development is crucial for optimizing procurement and sales strategies, improving market efficiency, and ensuring the

reliable operation of the power system.

Currently, the main electricity price forecasting methods used by scholars include statistical model-based prediction, machine learning model-based prediction, and deep learning model-based prediction. To provide a more comprehensive overview, a summary Table 1 is added below, which compares these methods in terms of advantages, limitations, and representative models. Statistical and machine learning-based approaches analyze historical electricity price data for a specific region and establish models for forecasting. Traditional statistical models—such as linear regression [5], ARIMA [6], exponential smoothing [7], and Kalman filtering [8]. These methods are valued for their interpretability and computational efficiency, making them suitable for relatively simple and stable time series. However, electricity price data often exhibit complex features such as volatility clustering, structural breaks, which challenge the performance of linear and stationary models. To address these characteristics, more advanced statistical learning techniques—such as GARCH (Generalized Autoregressive Conditional Heteroskedasticity) models [9] and regime-switching models [10]—have been widely adopted. For instance, GARCH models effectively model volatility clustering, while regime-switching models can detect shifts in market states (e.g., high- and low-price regimes), thereby improving forecast accuracy in dynamic environments. Despite

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their strengths, these statistical models often rely on strong parametric assumptions and may struggle with high-dimensional or irregularly sampled data.

Machine learning-based methods, such as artificial neural networks (ANN) [11], support vector machines (SVM) [12], and random forests [13], exhibit strong fitting capabilities. However, machine learning methods often struggle to handle data with significant nonlinearity and volatility and have difficulty capturing dependencies between data points [14].

Compared to previous approaches, deep learning-based methods offer superior accuracy and generalization capabilities. They automatically learn features from data, effectively capturing the nonlinear characteristics and temporal dependencies of electricity prices. This capability has driven increasing scholarly adoption for electricity price forecasting. Recurrent neural network (RNN)-based architectures exemplify this trend, including models such as Long Short-Term Memory (LSTM) [15,16], Bidirectional LSTM (BiLSTM) [17], Gated Recurrent Unit (GRU) [18], and Temporal Convolutional Network (TCN) [19]. While these models exhibit strong adaptability to complex data patterns, they struggle with gradient vanishing and long-term dependency capture in long-sequence data.

In recent years, Transformers have emerged as the dominant forecasting framework due to their superior performance. Scholars have developed variants like Informer, Crossformer, iTransformer, and TimeXer. Unlike RNN-based models, Transformers leverage attention mechanisms to effectively capture long-sequence dependencies, overcoming gradient vanishing, and their parallel computing capability significantly enhances training efficiency [20]. Literature [21] proposes a deep neural network integrating Transformer with Time2Vec layers, using probabilistic forecasting for 8-hour-ahead electricity price prediction and achieving notable gains across datasets and benchmarks. Literature [22] introduces a regional forecasting method combining Transformer and Graph Neural Networks (GNN): Transformers extract temporal features, while GNNs capture spatial dependencies across regions, offering new insights for large-scale power market estimation.

Although the self-attention mechanism in Transformer models excels at capturing long-term dependencies, its performance often deteriorates when handling non-standard or irregular time periods. For instance, during holidays, weekends, or periods of sharp price fluctuations, electricity prices significantly deviate from typical patterns, making accurate modeling challenging. Moreover, existing time-series modeling approaches largely overlook the explicit consideration of special periods at the model encoding level, failing to fully capture their structural impact across different periods. These limitations highlight the need for more robust and adaptive modeling frameworks, thereby motivating the contributions of our work.

Deep learning models have effectively captured complex temporal dependencies in time series, but the integration of endogenous and exogenous variables remains underexplored [23]. Endogenous variables are the target sequences to be predicted, while exogenous variables

serve as covariates that improve forecasting accuracy. In electricity price forecasting, incorporating external factors—such as climate, policy, economic activity, supply-demand dynamics, and holiday effects—can significantly enhance performance, especially when these drivers cause system fluctuations. Future research should focus on integrating multi-dimensional data, including climate, supply, demand, and holiday patterns, to achieve more accurate and comprehensive predictions.

Aiming at the limitations of the aforementioned electricity price prediction model, this paper proposes a novel short-term electricity price forecasting framework—HE-DTimeXer. The main contributions of this study are as follows:

- 1) **Exogenous Variable Analysis:** As TimeXer is designed for exogenous variable integration, we use PCC and MIC-TSA to identify significant factors (e.g., load, temperature, wind speed) and jointly model their impact, leveraging the model's native ability to capture collective external influences. Significant variables are selected for joint modeling to capture their collective impact.
- 2) **Holiday Encoding Mechanism:** To address distinct price patterns during holidays, we enhance the TimeXer embedding layer by introducing a Holiday Encoding (HE) mechanism. This mechanism encodes the influence of different holidays into the input representation, enabling the model to better adapt to special holiday scenarios and improve forecasting accuracy in such periods.
- 3) **Differential Attention Mechanism:** To mitigate the limitations of self-attention and handle noise or abnormal fluctuations in electricity price data, we replace the self-attention in TimeXer with a Differential Attention mechanism. This captures temporal dependencies in the electricity price series, reducing accuracy loss caused by noise.

## 2. Relevant methods

The workflow begins with data preprocessing of electricity prices and exogenous variables, followed by correlation analysis and dataset splitting. The proposed HE-DTimeXer model, incorporating holiday encoding, differential and cross-attention mechanisms, and normalization, is trained on the training set and validated on the validation set. Hyperparameters are adjusted based on validation performance, and the optimized model is used to generate prediction results on the test set. The entire process is summarized in the flowchart (Fig. 1).

### 2.1. Correlation analysis

To identify influential exogenous variables, we adopt a dual approach by analyzing both linear and nonlinear relationships. We use the Pearson Correlation Coefficient (PCC) to detect linear dependencies and the Maximal Information Coefficient with Time Series Analysis (MIC-TSA) to capture nonlinear and time-lagged correlations. This combined framework ensures a more comprehensive selection of relevant factors—such as load, temperature, and wind speed—enabling

**Table 1**  
Summary and comparison of electricity price forecasting methods.

| Category                          | Models                                              | Advantages                                                                                                           | Limitations                                                                                          |
|-----------------------------------|-----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| Statistical Models                | Linear Regression, ARIMA<br>GARCH, Kalman filtering | Simple structure, fast training, high interpretability                                                               | Limited ability to capture non-linear patterns; poor performance on non-stationary data              |
| Machine Learning                  | ANN                                                 | Strong non-linear fitting capability; suitable for complex patterns                                                  | Highly dependent on feature engineering; poor long-term dependency modeling; prone to overfitting    |
| Deep Learning (RNN-based)         | SVM                                                 | Effective in capturing temporal dependencies; suitable for short-term forecasting                                    | Suffers from vanishing/exploding gradients; limited long-range dependency modeling                   |
|                                   | Random Forest                                       |                                                                                                                      |                                                                                                      |
|                                   | LSTM                                                |                                                                                                                      |                                                                                                      |
|                                   | GRU                                                 |                                                                                                                      |                                                                                                      |
| Deep Learning (Transformer-based) | BiLSTM                                              | Efficient long-range dependency modeling via self-attention; supports parallel computation; high prediction accuracy | High model complexity; requires large amounts of data; Poor adaptability of self attention mechanism |
|                                   | BiGRU                                               |                                                                                                                      |                                                                                                      |
|                                   | Transformer, Informer, Crossformer, TimeXer         |                                                                                                                      |                                                                                                      |

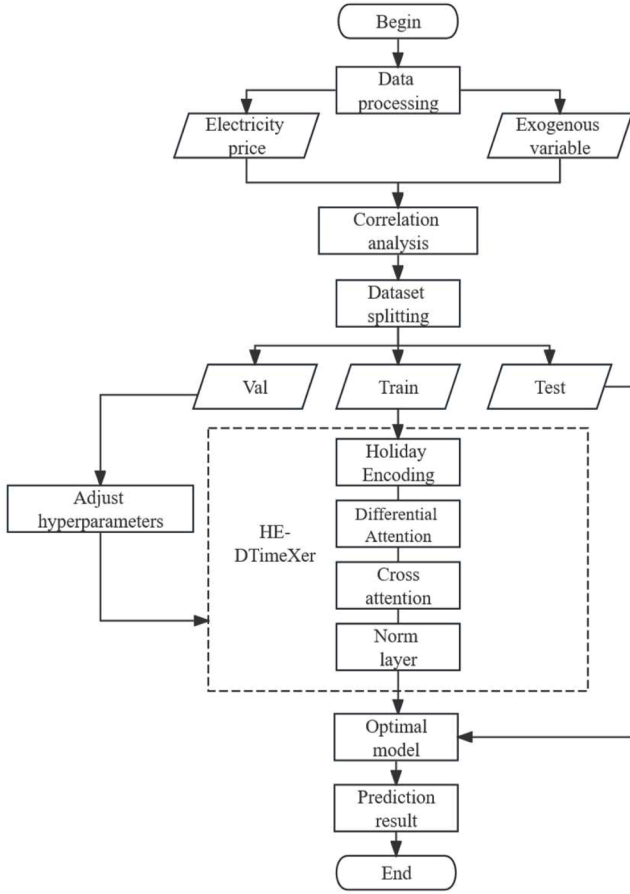


Fig. 1. Overall flowchart.

robust joint modeling of their collective impact on electricity prices.

### 2.1.1. PCC correlation analysis

The Pearson Correlation Coefficient (PCC) is a commonly used statistical method for analyzing the linear relationship between two variables. The formula is as follows:

$$R = \frac{\sum (X - \bar{X})(Y - \bar{Y})}{\sqrt{\sum (X - \bar{X})^2 \sum (Y - \bar{Y})^2}} \quad (1)$$

Where  $R$  is the calculated PCC correlation coefficient;  $X$  and  $Y$  represent the variables;  $\bar{X}$  and  $\bar{Y}$  are the mean values of  $X$  and  $Y$ , respectively.

### 2.1.2. MIC-TSA correlation analysis

The Maximal Information Coefficient (MIC) is a method for exploring the correlation of nonlinear relationships, which does not rely on assumptions about data distribution. It identifies complex nonlinear relationships and quantifies their strength using a shared information measure. However, conventional MIC fails to detect causal relationships across different time lags. Therefore, this paper innovatively introduces a causal time-lagged index  $\tau$  to measure the causal correlation between electricity price sequence variables  $X(t)$  and  $Y(t+\tau)$  under different time lags, enhancing sensitivity to time delays and temporal structures. By calculating MIC values across multiple time lag windows and selecting the maximum value as the final result, it characterizes the maximum linear or nonlinear relationship between the two time series under different time lags. The formula is as follows:

$$MIC_\tau = \text{MAX}(I(X(t), Y(t+\tau)) / \ln_{\min}(|X(t)|, |Y(t+\tau)|)) \quad (2)$$

where  $I(X(t), Y(t+\tau))$  represents mutual information, and  $|X(t)|, |Y(t+\tau)|$  are sample sizes.

### 2.1.3. Correlation analysis results

Considering the significant differences in the impact of exogenous variables in different seasons, this paper divides the dataset by season, conducts correlation analysis between electricity prices and other exogenous variables, and selects key features for modeling together.

The results of the correlation analysis are shown in Tables 2 and 3. Using a weighted fusion approach to combine the correlation analyses from PCC and MIC-TSA, the fusion process is as follows:

$$\text{Score} = \alpha|\text{MIC} - \text{TSA}| + (1 - \alpha)|\text{PCC}| \quad (3)$$

Considering that nonlinearity and lagged effects are more prominent in electricity price, a higher weight (0.6) is assigned to MIC-TSA. Ultimately, this paper selects the top 4–5 exogenous variables with the highest impact on electricity prices, together with electricity, as input features, as shown in Table 4.

## 2.2. Holiday encoding

Conventional time-series models conflate holidays with regular weekdays, overlooking the unique impact of holidays on electricity demand and price fluctuations. This underestimates their strong influence, compromising prediction accuracy. To address this, we propose Holiday Encoding (HE), which embeds holiday influences via importance weights. Assigning distinct weights to holidays directs model attention to specific periods, enhancing prediction accuracy for these segments. Using a custom holiday list, we generate a label column for each timestep (0: non-holiday; 1: regular holiday/weekend; 2: special holiday) and attach corresponding weights  $H$ :

$$H = \alpha(t)N_{\text{normal}} \sum_{i \in \text{holiday}} L_i / N_{\text{holiday}}^H \sum_{i \in \text{normal}} L_i \quad (4)$$

Where  $L_i$  is the load for  $i$  time steps, and  $N_{\text{normal}}$  and  $N_{\text{holiday}}^H$  are the number of weekdays and class  $H$  holiday days, respectively.

A time decay smoothing hyperbolic function  $\alpha(t)$  is designed to address long-term holiday characteristics. Short-term holidays are assigned a value of 1, while long-term holidays are calculated using this function, enabling gradual weight recovery around the holiday period instead of abrupt transitions. The formula is provided below:

$$\alpha(t) = H_t + H_{\text{base}} - H_t / 1 + \exp(k \cdot (t - t_{\text{holiday}})) \quad (5)$$

Where  $t_{\text{holiday}}$  is the long-term holiday center time step,  $H_t$  is the long-term holiday center time step weight,  $H_{\text{base}}$  is the weekday weight taken as 1, and  $k$  is the smoothing index.

The holiday encoding process is illustrated in Fig. 2. The linear projection matrix  $W_p$ , positional encoding  $W_{\text{pos}}$ , and holiday encoding  $W_{\text{holiday}}$  are applied to  $i$  slices  $x_p^i$ , serving as the input for the next layer:

$$P^i = W_p x_p^i + W_{\text{pos}} + W_{\text{holiday}} \quad (6)$$

**Table 2**  
PCC correlation analysis results.

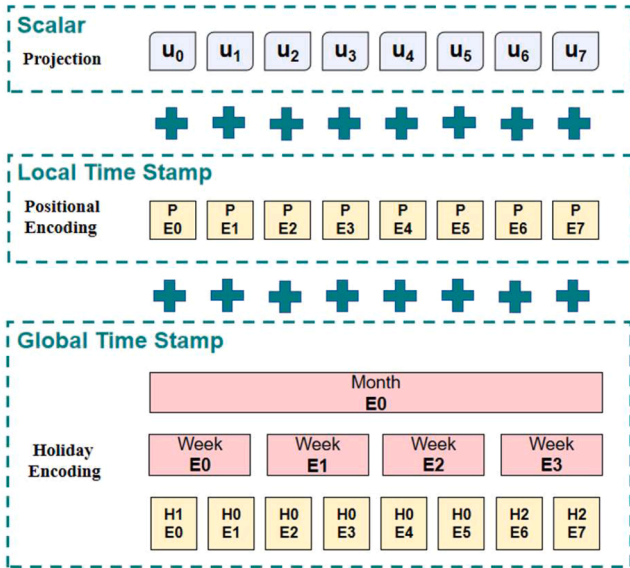
| Exogenous Variables | Spring | Summer | Autumn | Winter |
|---------------------|--------|--------|--------|--------|
| Wind Speed          | 0.043  | 0.028  | 0.011  | 0.003  |
| Radiation           | 0.231  | 0.477  | 0.292  | 0.383  |
| Humidity            | −0.142 | −0.264 | −0.331 | −0.102 |
| Temperature         | 0.341  | 0.692  | 0.434  | 0.306  |
| Precipitation       | 0.354  | 0.402  | 0.337  | 0.313  |
| Air Pressure        | −0.073 | 0.022  | −0.044 | −0.037 |
| Total Load          | 0.674  | 0.884  | 0.731  | 0.783  |
| Power Generation    | −0.539 | −0.672 | −0.534 | −0.611 |
| Holidays            | 0.663  | 0.651  | 0.603  | 0.579  |

**Table 3**  
MIC-TSA correlation analysis results.

| Exogenous Variables | Spring | Summer | Autumn | Winter |
|---------------------|--------|--------|--------|--------|
| Wind Speed          | 0.024  | 0.077  | 0.082  | 0.013  |
| Radiation           | 0.201  | 0.375  | 0.173  | 0.266  |
| Humidity            | 0.002  | 0.094  | 0.114  | 0.057  |
| Temperature         | 0.332  | 0.379  | 0.297  | 0.228  |
| Precipitation       | 0.247  | 0.259  | 0.324  | 0.298  |
| Air Pressure        | 0.094  | 0.012  | 0.034  | 0.077  |
| Total Load          | 0.643  | 0.687  | 0.596  | 0.574  |
| Power Generation    | 0.535  | 0.516  | 0.487  | 0.501  |
| Holidays            | 0.459  | 0.443  | 0.411  | 0.417  |

**Table 4**  
Feature selection results.

|                     |                                                                               |
|---------------------|-------------------------------------------------------------------------------|
| Spring              | Precipitation, Temperature, Total Load, Power Generation, Holidays            |
| Summer              | Radiation, Temperature, Total Load, Power Generation, Holidays                |
| Autumn              | Humidity, Temperature, Total Load, Power Generation, Holidays                 |
| Winter              | Precipitation, Temperature, Total Load, Power Generation, Holidays            |
| throughout the year | Radiation, Temperature, Precipitation, Total Load, Power Generation, Holidays |



**Fig. 2.** Holiday encoding format.

### 2.3. HE-DTimeXer

TimeXer is a model based on the Transformer architecture, whose core mechanism captures the temporal dependencies within endogenous variables and the dependencies among exogenous variables through self-attention and cross-attention mechanisms. The overall architecture consists of an embedding layer, a time-dependent extraction layer, and a prediction layer, as shown in Fig. 3.

#### 2.3.1. Embedding layer

For a given endogenous variable  $x_{1:T} = \{x_1, x_2, x_3, \dots, x_T\}$  and C exogenous variables  $z_{1:T} = \{z_{1:T}^1, z_{1:T}^2, z_{1:T}^3, \dots, z_{1:T}^C\}$ , TimeXer employs a patch-based representation approach, dividing the endogenous sequence into non-overlapping patches and projecting each into a temporal token. Due to granularity differences between endogenous and exogenous variables, joint modeling at disparate granularities may cause information mismatch. To address this, the model introduces a

learnable global token, enabling exogenous variables to participate in endogenous representation learning from a macroscopic perspective, thereby bridging the causal information gap between exogenous variables and endogenous time segments:

$$\{p_1, p_2, p_3, \dots, p_N\} = \text{Patch}(x) \quad (7)$$

$$P_{en} = \text{HolidayEncoding}(p_1, p_2, p_3, \dots, p_N) \quad (8)$$

$$G_{en} = \text{Learnable}(x) \quad (9)$$

In the formula, let L be the length of a patch,  $N = \lfloor T/L \rfloor$  represents the number of patches divided, HolidayEncoding refers to the holiday encoding process described above, and finally, the N patch fragments are embedded into  $P_{en}$  and  $G_{en}$ , which are then input into the encoder together.

Exogenous variables primarily serve to facilitate accurate prediction of endogenous variables. Although patch-level representation has finer granularity, it often introduces unnecessary computational complexity and may not yield better performance. Therefore, different exogenous variables are more suitable for modeling in a variable-level manner:

$$V_{ex,j} = \text{VariateEmbed}(z^j), i \in \{1, \dots, C\} \quad (10)$$

#### 2.3.2. Dependency extraction layer

The original Timexer model employs self-attention and cross-attention mechanisms to capture the temporal dependencies of endogenous and exogenous variables. However, self-attention treats all time points equally and computes the correlation between each element and others, which may cause the model to overemphasize noise and thus reduce generalization ability [24,25]. In electricity price forecasting, strong noise often arises from sudden events, market fluctuations, and supply-demand changes, further exacerbating this issue. To address this, we introduce a differential attention mechanism that more effectively captures temporal dependencies in endogenous variables while mitigating the adverse impact of noise on prediction accuracy.

The Differential Attention mechanism calculates the difference between two input signals to eliminate common-mode noise (i.e., interference from identical parts of the input signals). It treats the two softmax attention weights as two perspectives of the input features, and their difference can eliminate the influence of irrelevant features, thereby focusing more on useful information. This mechanism reduces the impact of noise on the model, enhancing its stability and accuracy. The overall architecture is shown in Fig. 4.

Firstly, the mapped vectors  $P_{en}^l$  provided by the embedding layer are projected into query, key, and value vectors  $Q_1, Q_2, K_1, K_2, V$ . Then, the output of the differential attention is calculated as follows:

$$[Q_1 : Q_2] = P_{en}^l W^Q, [K_1 : K_2] = P_{en}^l W^K, V = P_{en}^l W^V \quad (11)$$

$$\text{DiffAtten}(Q, K, V) = \left( \text{Softmax}\left(Q_1 K_1 / \sqrt{d}\right) - \mu \text{Softmax}\left(Q_2 K_2 / \sqrt{d}\right) \right) V \quad (12)$$

$$\hat{P}_{en}^l = \text{LayerNorm}(P_{en}^l + \text{DiffAtten}(P_{en}^l)) \quad (13)$$

In the equations,  $l$  represents the  $l$ -th attention layer, and  $\mu$  is a learnable scalar that is continuously adjusted during training via normalization.

Next, the cross-attention layer is applied to multimodal learning tasks to model dependencies between different modalities. In TimeXer, the cross-attention layer models the dependencies between exogenous and endogenous variables by taking exogenous variables as keys and endogenous variable representations as queries. Using the learnable global token  $G_{en}^l$  mentioned above as an exchange medium, the specific process is as follows:

$$\hat{G}_{en}^l = \text{LayerNorm}(G_{en}^l + \text{CrossAttention}(G_{en}^l, V_{ex})) \quad (14)$$

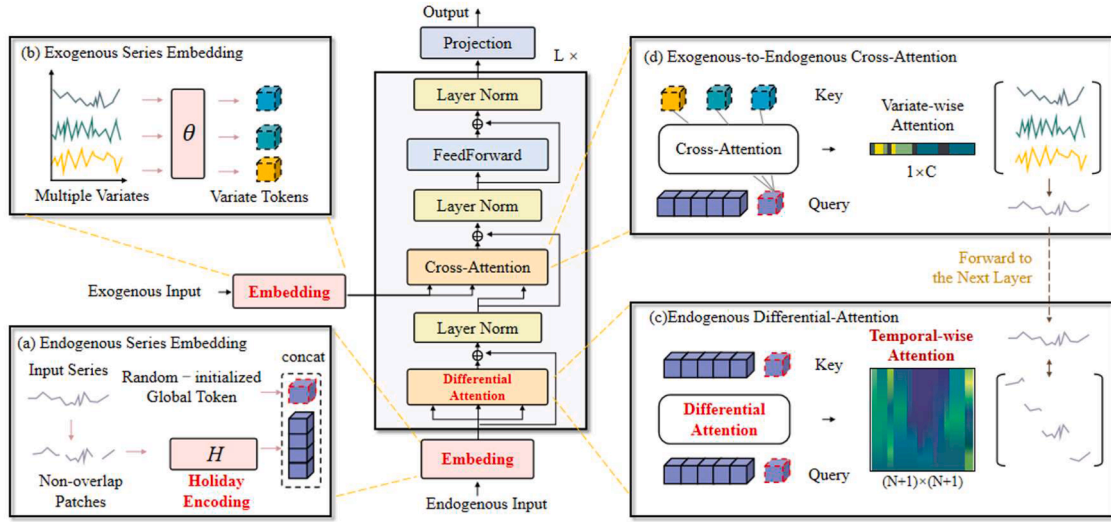


Fig. 3. The overall architecture of HE-DTimeXer mode.

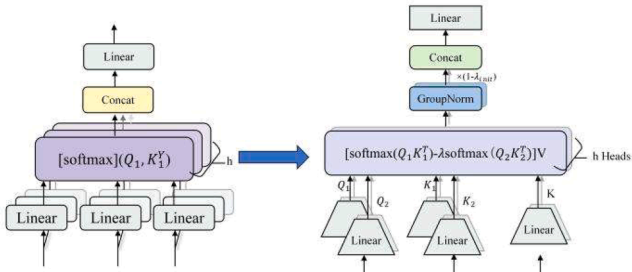


Fig. 4. The overall architecture of Differential Attention.

### 2.3.3. Dependency extraction layer

Finally, the two results from the dependency extraction layer are flattened and passed to the projection layer, where the high-dimensional features extracted by the model are mapped to specific prediction values via the weight matrix  $WW$  and bias  $bb$ . The projection layer mapping formula is as follows:

$$y^i = W[\hat{P}_{en}^i, \hat{G}_{en}^i] + b \quad (15)$$

## 3. Experimental analysis

To verify the effectiveness of the proposed model, electricity price datasets from different regions are employed for multi-dimensional experiments.

**Conventional Dataset:** Actual price data from Shandong Province, China (2023–2024, 30-minute interval), including meteorological features, energy generation, and load. This dataset is used to assess performance in conventional forecasting scenarios and serves as the primary dataset.

**Generalization Dataset:** Actual price data from the German market (2017–2020) and the Belgian market (2016–2018), both with 1-hour intervals, incorporating energy generation and load. These datasets, characterized by high volatility, are used to evaluate robustness and generalization under noisy conditions.

In this paper, min-max normalization is used to scale the data to  $[0,1]$ . As shown in Eq:

$$X_{norm} = \frac{x - x_{min}}{x_{max} - x_{min}} \quad (16)$$

Segmentation is done only on the training set using calendar features

(e.g., weekday, holiday) from timestamps. It does not use electricity prices or future data. The same segmentation is applied during inference based on timestamp. All preprocessing, including segmentation, is performed only on training data, avoiding any data leakage.

### 3.1. Experimental setup

The algorithmic models were implemented in Python 3.9 and developed based on the PyTorch framework. The final hyperparameter settings of the model are shown in the Table 5 below:

### 3.2. Evaluation metrics

In this paper, Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and Mean Absolute Percentage Error (MAPE) are used as metrics to assess the model prediction accuracy.

$$MAE = \frac{1}{N} \sum_{i=1}^N |\hat{y}_i - y_i| \quad (17)$$

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (\hat{y}_i - y_i)^2} \quad (18)$$

$$MAPE = \frac{100\%}{N} \sum_{i=1}^N \frac{|\hat{y}_i - y_i|}{y_i} \quad (19)$$

Where  $y_i$  is the actual value of the test sample,  $\hat{y}_i$  is the predicted value of the model, and  $\bar{y}_i$  is the summed average of the actual values of the sample.

**Table 5**  
Hyperparameter settings.

| Hyperparameter              | Value | Hyperparameter        | Value            |
|-----------------------------|-------|-----------------------|------------------|
| Prediction window           | 96    | Dropout rate          | 0.1              |
| Prediction step             | 4     | Batch size            | 64               |
| Hidden layer dimension      | 128   | training epochs       | 100              |
| Encoder-decoder layers      | 128   | Initial learning rate | 0.0001           |
| Patch length                | 16    | Early stopping        | 0.2              |
| Attention Layer             | 4     | Validation method     | cross-validation |
| Train/Validation/Test ratio | 7:1:2 | Optimizer             | Adam             |

### 3.3. Analysis of exogenous variable importance

To verify the importance of exogenous variables, we conduct prediction experiments using HE-DTimeXer with and without exogenous variable sequences (using only the electricity price sequence), and compare the two scenarios by calculating the MAE and RMSE metrics.

As shown in the experimental metrics in Table 6, the model incorporating exogenous variables outperforms the model without them across all seasons and the entire year. This is evidenced by lower values in evaluation metrics such as MSE, MAE, and MAPE. Particularly in seasons with significant electricity price fluctuations—such as summer and winter—the inclusion of exogenous variables allows the model to better capture external factors influencing electricity prices. Consequently, predictive performance is further enhanced. The experimental results demonstrate that exogenous variables play a crucial role in improving prediction accuracy.

### 3.4. Analysis of exogenous variable contributions

To systematically evaluate the contribution of different exogenous variables to the prediction results, this study incorporates SHAP analysis (Shapley Additive Explanations) into the HE-DTimeXer model. SHAP quantifies the contribution of each exogenous variable to the model's predictions by assigning a Shapley value to each input feature, thereby measuring its influence on the final output. From Figs. 5–8, The SHAP plots across different seasons reveal significant variations in the contributions of the exogenous variables. Load emerges as the most influential factor, exhibiting high SHAP values and substantial variability across seasons, indicating its dominant role in the model. Generation and holiday indicators follow, with moderate SHAP values and notable importance. In contrast, climate features (e.g., temperature, precipitation) show lower SHAP values, suggesting limited direct impact. However, their influence, though indirect via electricity demand, remains non-negligible, particularly under extreme weather conditions.

### 3.5. Comparison and analysis of different prediction models

To validate the accuracy of the HE-DTimeXer prediction model in this paper, it is compared with two classic statistical learning models (ARIMAX, GARCH), two machine learning models (SVM, XGBOOST), and multiple mainstream deep learning models in recent years. Meanwhile, to more clearly compare the performance of different forecasting models, the MAE, RMSE, and MAPE metrics of each model are calculated.

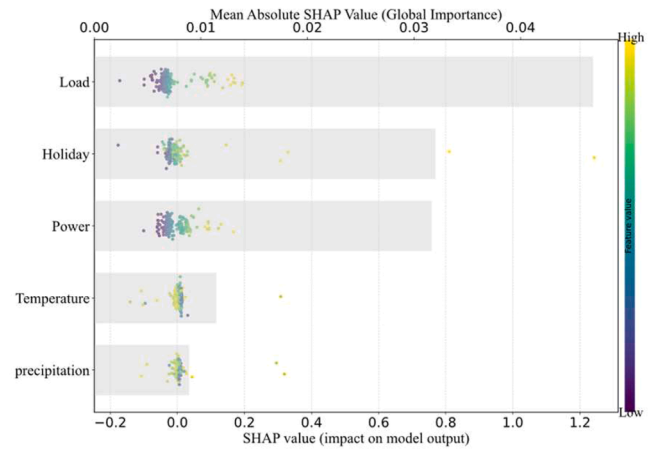
From Fig. 9, the prediction results clearly demonstrate that statistical learning models and traditional machine learning models perform significantly worse than deep learning models, primarily due to their inherent limitations in handling the complex nonlinear characteristics and temporal dependencies of electricity price time series.

As illustrated in Fig. 10, the prediction results reveal distinct differences in performance among the models. Traditional recurrent neural networks, such as LSTM, exhibit notable limitations in electricity price

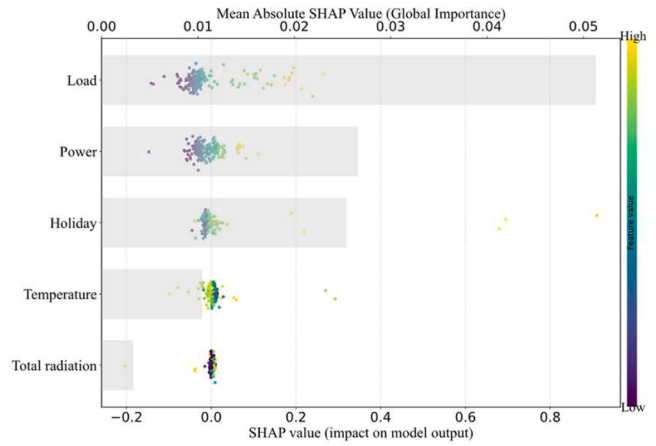
**Table 6**

Experimental indicators of the importance of exogenous variables.

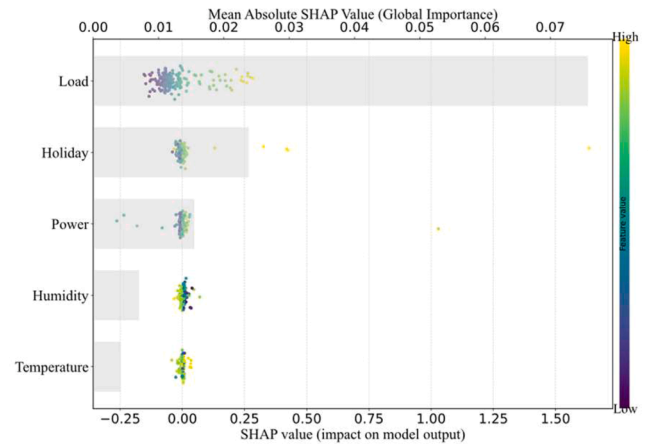
| season         | Exogenous Variable | RMSE  | MAE   | MAPE  |
|----------------|--------------------|-------|-------|-------|
| Spring         | ✓                  | 0.171 | 0.124 | 3.737 |
|                | ×                  | 0.179 | 0.129 | 3.941 |
| Summer         | ✓                  | 0.201 | 0.144 | 4.771 |
|                | ×                  | 0.221 | 0.156 | 5.214 |
| Autumn         | ✓                  | 0.167 | 0.119 | 3.650 |
|                | ×                  | 0.174 | 0.126 | 3.665 |
| Winter         | ✓                  | 0.194 | 0.138 | 4.425 |
|                | ×                  | 0.207 | 0.146 | 4.879 |
| The whole year | ✓                  | 0.159 | 0.103 | 3.323 |
|                | ×                  | 0.171 | 0.117 | 3.514 |



**Fig. 5.** SHAP analysis results (spring).



**Fig. 6.** SHAP analysis results (summer).



**Fig. 7.** SHAP analysis results (autumn).

forecasting. Although they are capable of capturing long-term temporal dependencies, their sequential processing mechanism inherently introduces delays in responding to sudden short-term fluctuations. This leads to significant prediction lags and elevated error rates, particularly during periods of rapid price changes. In contrast, conventional Transformer-based models demonstrate improved performance by capturing global contextual information across long sequences, thereby mitigating the issue of temporal lag to a certain extent. However, these

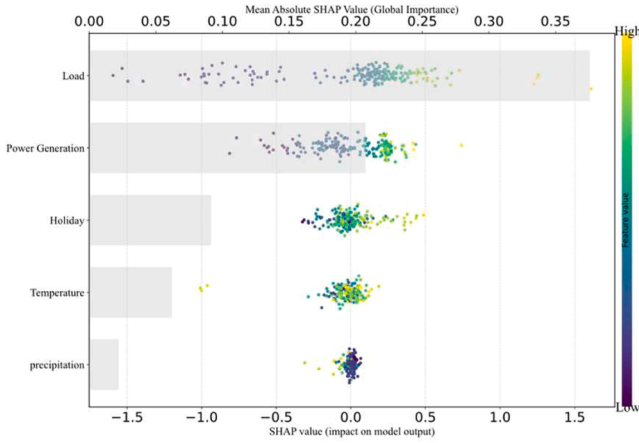


Fig. 8. SHAP analysis results (winter).

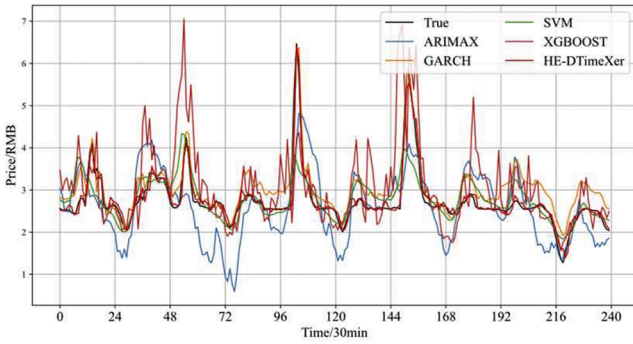


Fig. 9. Comparison Results of Statistical Learning and Machine Learning Models on Standard Dataset(left).

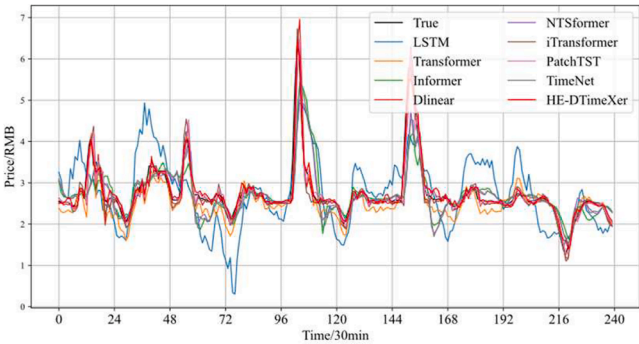


Fig. 10. Comparison results of deep learning models on the standard dataset(right).

models still struggle to accurately model fine-grained short-term dynamics, such as sharp peaks and troughs in electricity prices, due to their focus on long-range dependencies at the expense of local detail.

The proposed HE-DTimeXer model, by contrast, achieves the most accurate predictions across all evaluation metrics. This is primarily attributed to its enhanced noise adaptation capability and advanced feature extraction mechanism, which enable the model to effectively capture both long-term trends and short-term fluctuations. The integration of hybrid temporal modeling and dynamic attention further improves responsiveness to sudden market changes, resulting in superior predictive accuracy and robustness. As shown in Table 7, HE-DTimeXer significantly outperforms deep learning baseline models, with RMSE reduced by 22.81 %~71.74 %, MAE by 17.62 %~78.71 %, and MAPE by 22.01 %~77.09 %.

Table 7

Results of different predictive model indicators(Conventional Dataset).

| Model        | RMSE         | MAE          | MAPE         |
|--------------|--------------|--------------|--------------|
| ARIMAX       | 0.801        | 0.617        | 19.977       |
| XGBOOST      | 0.729        | 0.541        | 17.479       |
| SVM          | 0.710        | 0.556        | 17.331       |
| LSTM         | 0.625        | 0.484        | 15.033       |
| GARCH        | 0.529        | 0.403        | 12.681       |
| Transformer  | 0.481        | 0.295        | 10.067       |
| TimeNet      | 0.440        | 0.261        | 8.862        |
| NTSformer    | 0.433        | 0.256        | 8.615        |
| Informer     | 0.260        | 0.201        | 7.437        |
| Dlinear      | 0.241        | 0.136        | 4.566        |
| iTransformer | 0.233        | 0.121        | 3.823        |
| PatchTST     | 0.206        | 0.125        | 4.261        |
| HE-DTimeXer  | <b>0.159</b> | <b>0.103</b> | <b>3.323</b> |

and MAPE by 22.01 %~77.09 %.

### 3.6. Generalization experiment analysis

To evaluate the robustness and generalization ability of the proposed HE-DTimeXer model in high-noise environments, this study selects two datasets with significant volatility. Visualization analysis of regular weekdays in the test set is also conducted.

Figs. 11–14 display the fitting performance of various models on two generalization datasets. Clearly, statistical learning and traditional machine learning models exhibit significant deviations on the two electricity price generalization datasets characterized by high volatility, even failing to capture the basic trend. Mainstream deep learning models, while performing better, still show some errors—for instance, their performance is slightly worse at peaks and troughs due to the influence of noise. However, the proposed HE-DTimeXer model accurately captures the fluctuation trends of electricity prices. Particularly during periods of severe market volatility, its prediction curve closely aligns with the actual price movement—demonstrating strong noise adaptation ability, flexible response to sudden market fluctuations, and precise capture of short-term price changes. As shown in Table 8, the metrics of HE-DTimeXer outperform those of other models across all datasets.

To more intuitively demonstrate the performance differences of the model, this article supplements the error scatter plot and histogram (Fig. 15) and box plot (Fig. 16). From the scatter plot in Fig. 15, it can be seen that HE-DTimeXer has the best consistency between predicted values and true values on the three datasets, with points closely distributed around the diagonal of  $y = x$ ; The histogram shows that the error distribution is the narrowest and concentrated around 0, indicating that the prediction error is small and stable.

From the box plot in Fig. 16, it can be seen that HE-DTimeXer has the most compact box plot, with the median of all three datasets close to 0, the smallest interquartile range (IQR), and the least outliers, indicating that its prediction error is the smallest and its stability is the highest; The

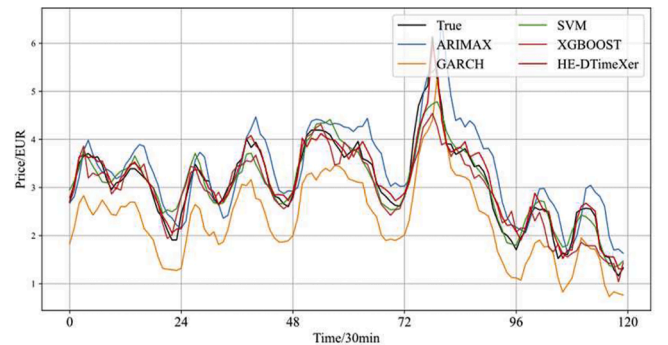


Fig. 11. Comparison Results of Statistical Learning and Machine Learning Models on Generalization dataset 1.

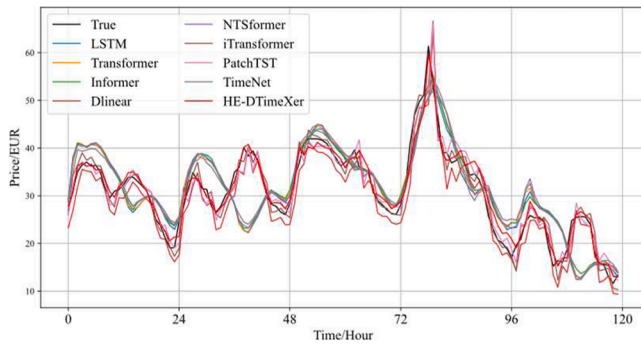


Fig. 12. Comparison results of deep learning models on Generalization dataset 1.

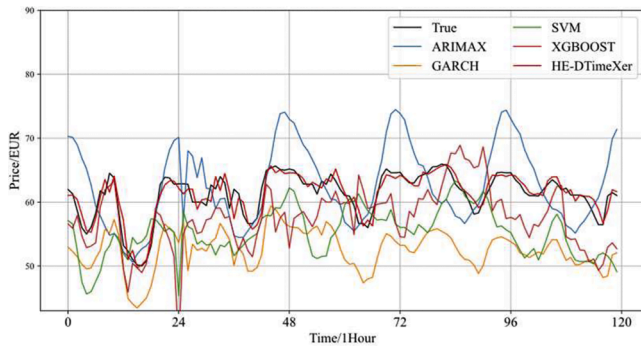


Fig. 13. Comparison Results of Statistical Learning and Machine Learning Models on Generalization dataset 2.

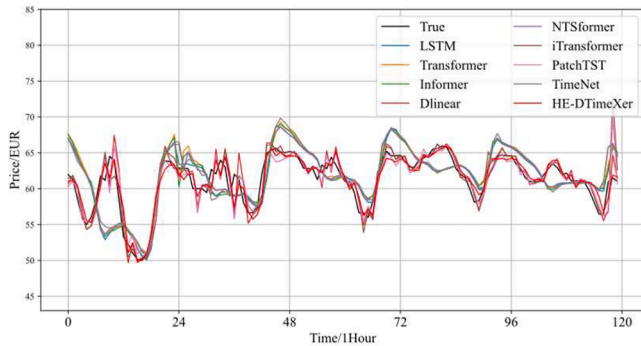


Fig. 14. Comparison results of deep learning models on Generalization dataset 2.

Table 8

Results of different predictive model indicators(Generalization Dataset1–2).

| Model        | Dataset1 |       |        | Model        | Dataset2 |       |       |
|--------------|----------|-------|--------|--------------|----------|-------|-------|
|              | RMSE     | MAE   | MAPE   |              | RMSE     | MAE   | MAPE  |
| GATCH        | 8.207    | 6.008 | 19.412 | ARIMAX       | 8.277    | 6.795 | 8.847 |
| ARIMAX       | 8.014    | 5.972 | 19.370 | GATCH        | 7.614    | 6.220 | 8.312 |
| SVM          | 7.445    | 5.573 | 18.231 | SVM          | 7.477    | 6.147 | 8.308 |
| XGBOOST      | 6.757    | 5.061 | 17.301 | XGBOOST      | 5.271    | 4.096 | 6.284 |
| LSTM         | 5.686    | 4.492 | 15.903 | LSTM         | 3.100    | 2.503 | 4.093 |
| Transformer  | 5.591    | 4.322 | 15.224 | TimeNet      | 3.007    | 2.458 | 4.024 |
| Informer     | 4.991    | 3.953 | 13.981 | Transformer  | 2.859    | 2.324 | 3.809 |
| TimeNet      | 4.671    | 3.472 | 12.046 | NTSformer    | 2.561    | 2.010 | 3.391 |
| NTSformer    | 4.173    | 3.001 | 10.447 | Informer     | 2.344    | 1.761 | 3.004 |
| iTransformer | 3.327    | 2.626 | 9.001  | Dlinear      | 1.788    | 1.347 | 2.031 |
| Dlinear      | 2.772    | 2.017 | 6.911  | PatchTST     | 1.718    | 1.165 | 1.917 |
| PatchTST     | 2.569    | 1.800 | 6.343  | iTransformer | 1.616    | 1.185 | 1.889 |
| HE-DTimeXer  | 1.728    | 1.364 | 4.854  | HE-DTimeXer  | 1.304    | 1.007 | 1.664 |

boxplots of other models (benchmark models) are relatively more dispersed, with larger IQR and more outliers, resulting in greater fluctuations in prediction errors.

### 3.7. Analysis of ablation experiment

To verify the effectiveness of the modules designed in this article, we removed each module from the model and constructed a variant model for ablation experiments to evaluate the contribution of each module to the overall performance of the model. Select the time period from September 29, 2024 to October 9, 2024 (before and after National Day) in the test set for comparative analysis.

Fig. 17 shows the electricity price curves during long holidays. Prices are generally lower than on regular days, with more pronounced periodic fluctuations. Compared to TimeXer, the improved DTimeXer demonstrates superior performance in noise handling. The differential attention mechanism effectively suppresses common-mode noise, allowing the model to focus more accurately on meaningful signals, particularly in capturing peaks and troughs, leading to more precise fitting and stable predictions. As shown in Table 9, DTimeXer achieves a 13.4 % reduction in RMSE, 9.1 % in MAE, and 10.7 % in MAPE compared to TimeXer. Furthermore, by incorporating holiday encoding into DTimeXer, HE-DTimeXer better captures price dynamics during the middle of the holiday period, aligning more closely with real-world fluctuations. As evident from the figure, holiday encoding also enables a smoother transition back to normal weekday levels at the beginning and end of the holiday period. Compared to DTimeXer, HE-DTimeXer achieves further improvements, with reductions of 8.6 % in RMSE, 5.5 % in MAE, and 9.5 % in MAPE.

### 3.8. Analysis of significance test

To ensure the statistical reliability of our comparative results, we conduct paired *t*-tests and Wilcoxon signed-rank tests across all currently popular electricity price forecasting baseline models and datasets. These non-parametric and parametric tests are performed to assess whether the performance differences between the proposed HE-DTimeXer and the baseline models are statistically significant at the 0.05 level.

The Table 10 results show that the proposed HE DTimeXer model exhibits consistent superiority in both conventional and high volatility generalization scenarios, especially when compared with advanced models such as Transformer, Informer, iTransformer, NTSformer, and PatchTST, with significant performance differences and robustness. Although the differences compared to some simple baseline models may not be statistically significant in certain cases, the overall results fully validate the advantages of HE DTimeXer in prediction accuracy, robustness, and cross-dataset generalization ability. These findings

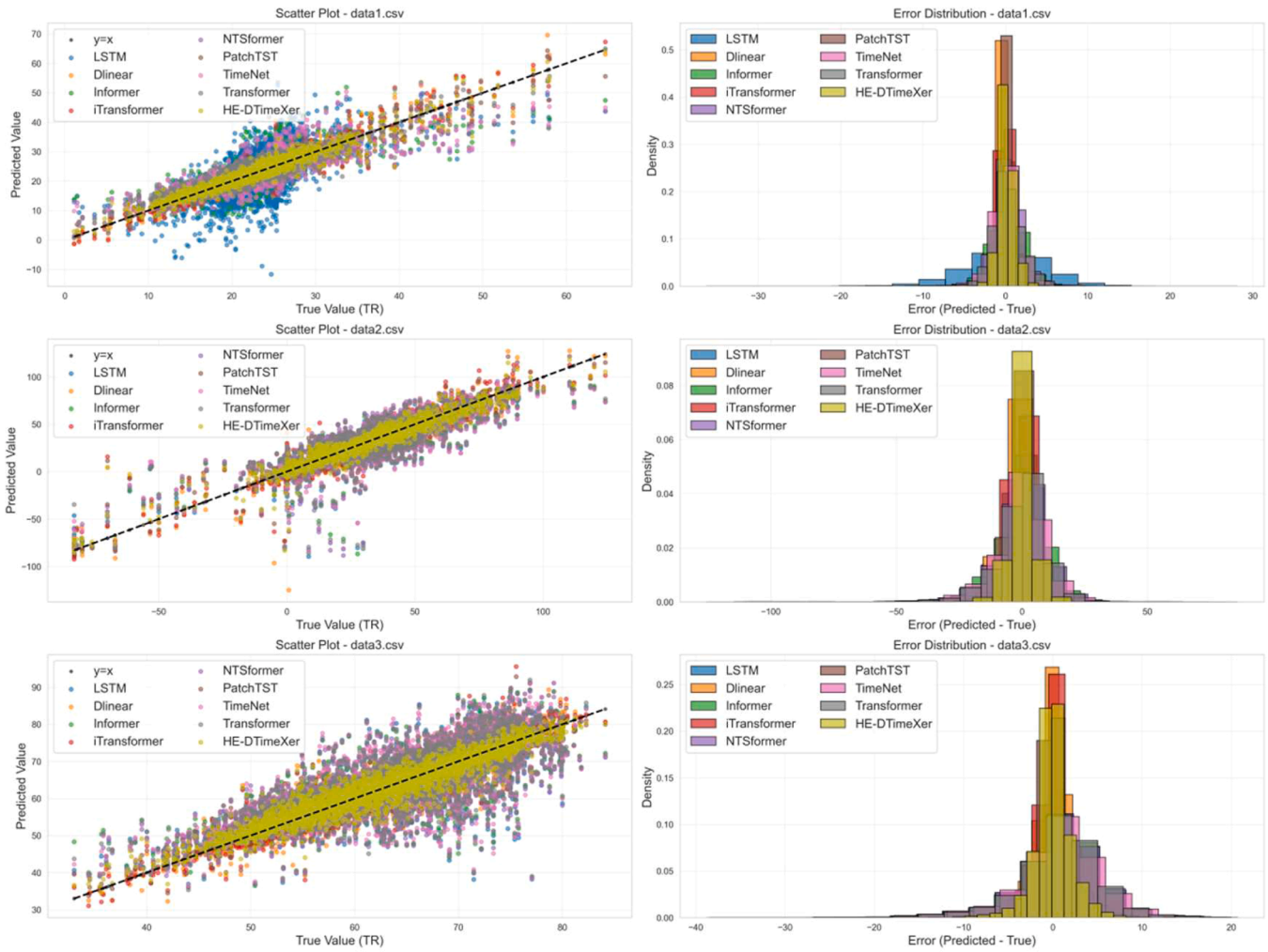


Fig. 15. Error Scatter Plots and Histogram of Deep Learning Models on three datasets.

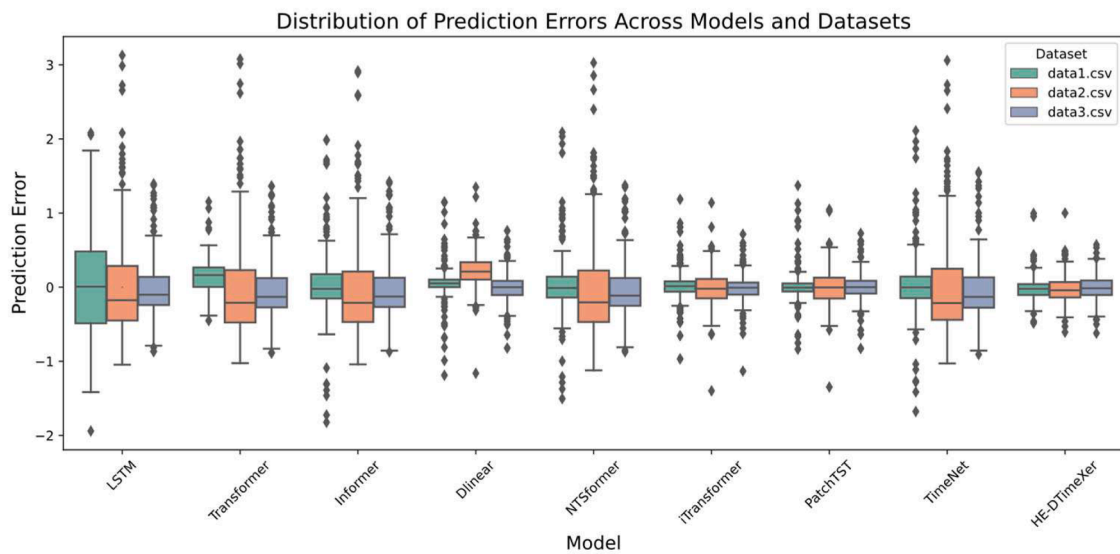


Fig. 16. Box plot of Deep Learning Models on three datasets.

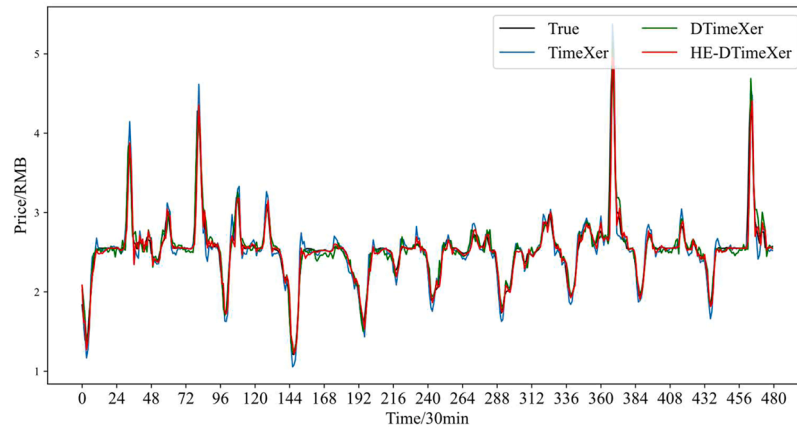


Fig. 17. Results of ablation experiment.

**Table 9**  
Results of ablation experiment metrics.

| Model       | RMSE         | MAE          | MAPE         | Time/s       |
|-------------|--------------|--------------|--------------|--------------|
| TimeXer     | 0.201        | 0.120        | 4.117        | 345.7        |
| DTimeXer    | 0.174        | 0.109        | 3.675        | 362.5        |
| HE-DTimeXer | <b>0.159</b> | <b>0.103</b> | <b>3.323</b> | <b>368.2</b> |

comprehensively demonstrate the effectiveness and practicality of HE DTimeXer in complex electricity market environments.

## Conclusion

This study addresses critical challenges in short-term electricity price forecasting—namely, the underutilization of exogenous variables (e.g., energy, load, climate, holidays) and the difficulties posed by data complexity, noise, and abnormal fluctuations. To this end, we propose the HE-DTimeXer model, a novel framework that integrates holiday encoding, correlation-driven feature selection, and a differential attention mechanism to enhance forecasting accuracy and robustness. The model effectively captures the dynamic interactions between electricity prices and external factors while improving noise suppression and peak-valley prediction.

Experimental results on multiple real-world datasets demonstrate the model's superior performance, with significant improvements in RMSE, MAE, and MAPE compared to state-of-the-art baselines. Notably, the differential attention mechanism enhances prediction quality during volatile periods, reducing error metrics by up to 13.4 % relative to the TimeXer baseline.

Beyond technical advancement, this work provides actionable insights for electricity market operators, supporting more efficient trading strategies and optimal power resource allocation. Future work will extend the framework to multi-market settings, exploring inter-market dependencies through multi-task learning and graph neural networks

to enable cross-market forecasting, and will also incorporate probabilistic forecasting to provide prediction intervals and improve uncertainty quantification.

## Glossary

N/A.

## Declaration of generative AI and AI-assisted technologies in the writing process

N/A.

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## Data availability

The data that support the findings of this study are available from the corresponding author, [TingHe], upon reasonable request.

## Declaration of competing interest

The authors declare no conflict of interest exists.

## CRediT authorship contribution statement

**Xunhui Li:** Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Ting He:** Writing – original draft, Software, Methodology, Conceptualization. **Wenlong Zhu:** Writing – review & editing, Validation, Investigation, Conceptualization. **Quan Xu:** Validation, Supervision, Investigation.

**Table 10**  
Results of *t*-test and Wilcoxon signed rank test.

| Dataset      | Conventional Dataset |          | Generalization Dataset 1 |          | Generalization Dataset 2 |          |
|--------------|----------------------|----------|--------------------------|----------|--------------------------|----------|
|              | <i>t</i> -tests      | Wilcoxon | <i>t</i> -tests          | Wilcoxon | <i>t</i> -tests          | Wilcoxon |
| LSTM         | 0.3724               | 0.3290   | 0.1847                   | 0.1308   | 0.3039                   | 0.0099   |
| Transformer  | 0.0000               | 0.0000   | 0.0602                   | 0.0259   | 0.1026                   | 0.0002   |
| Informer     | 0.0387               | 0.0124   | 0.1078                   | 0.0144   | 0.0329                   | 0.0063   |
| TimeNet      | 0.0284               | 0.1375   | 0.4155                   | 0.0308   | 0.1702                   | 0.0006   |
| NTSformer    | 0.0279               | 0.0089   | 0.0037                   | 0.0030   | 0.0197                   | 0.0026   |
| iTransformer | 0.0024               | 0.0005   | 0.0050                   | 0.0045   | 0.0045                   | 0.0017   |
| Dlinear      | 0.0000               | 0.0000   | 0.0000                   | 0.0000   | 0.0216                   | 0.0193   |
| PatchTST     | 0.0013               | 0.0000   | 0.0158                   | 0.0010   | 0.0086                   | 0.0029   |

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

Data will be made available on request.

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